BUSINESS ADMINISTRATION · BA 1204 · 2026–2027

Week 5 Practice Artifacts — Fillable

Business Performance: Map it, break it, fix it · OpenStax Ch. 10.1–10.6 · Artifact: current- and future-state process map with KPI, review point, escalation and fallback

Student name	Group	Company A	Company B
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Use this document if Miro is unavailable or lagging, or to tidy your maps after class. It is the same content as Miro Frames W5-1 to W5-4, plus a page to practise your A2 defence. Your current-state and future-state maps are the Application evidence for Weekly Record 5 and the process-map evidence in your A2 Video Essay.

HOW TO WORK IN THIS DOCUMENT (if Miro is slow)

1. **One copy per group.** One student downloads this file from Moodle and uploads it to your group chat in MS Teams (Files). Everyone opens it with "Open in browser" and types at the same time — just like Miro.
2. **Rename the file:** GROUP_CompanyB_Week5_Artifacts.docx (e.g., ITM2603_Kaspi_Week5_Artifacts.docx).
3. **Type in the empty white cells.** Cells grow as you type. Click a box ☐ to tick it.
4. **Your process map:** the step tables in Steps 2 and 4 ARE your map — you do not have to draw in Word. If you want a drawing, draw the swimlanes on paper, take a photo and insert it (Insert → Pictures).
5. **After class:** each student puts a screenshot of the finished tables into their own Weekly Record 5 (field 2). For Assignment 2, use the same screenshots in your video and Evidence file.
6. **You do not need to copy it into Miro later.** Working in this document counts the same as working in Miro.

Segment A — The Paper Airplane Factory

Teams of 5: Stations 1–4 + timekeeper/inspector. Customer order: 10 airplanes. Round 1 = batches of 5, push. Round 2 = batch of 1, pull (pass only when the next square is empty), inspect at every station.

Measure	Round 1 — Batch and push	Round 2 — One-piece flow	Change (R2 − R1)
Planes accepted in 3 minutes (throughput)	4	7	+3
Defects rejected by the inspector	2	1	−1
Planes still on desks at the stop (WIP)	8	2	−6
Time until the FIRST plane was accepted (lead time)	2:30	0:45	−1:45
Bottleneck station (where planes piled up)	Folding station	Inspection/hand-off	Bottleneck shifts

Debrief — answer in one or two sentences each

Question	Our answer
1 Which station was the bottleneck? What DATA shows it?	Folding was the bottleneck in the batch-and-push example: planes waited there while later stations were idle. Replace this with the station and counts observed by our group.
2 Which ONE number is the best KPI for this factory, and why?	Accepted planes per three minutes is the best output KPI because it measures completed good units; read it with defects so speed does not reduce quality.
3 Defects were found only at the end in Round 1. Where should the human review point be?	Place inspection immediately after the station where the defect is created, rather than waiting until final inspection.
4 A robot for ONE station: which one? What happens if you automate a station that is not the bottleneck?	Automate the bottleneck folding step only if it is the constraint. Automating a non-bottleneck may increase work-in-progress without raising throughput.
5 The robot breaks at minute 2. What is the manual fallback, and who decides to switch?	The timekeeper or inspector notices failure; the team lead stops the robot and switches to manual folding. Record downtime and resume one-piece flow.

No paper? Watch Lean Smarts, “Lean Manufacturing One Piece Flow vs. Mass Production Paper Airplane Simulation,” YouTube. https://www.youtube.com/watch?v=jPp_3-zfPaQ — and fill the table from the video.

Activity — Company B process map

Step 1: choose ONE Company B process

Item	Your answer
Process name (the same one you use in A2)	KMG domestic diesel order-to-loading process', 'A B2B order or seasonal demand forecast is received.
Trigger (what starts it)	Customer receives allocated fuel and loading documentation; process ends after loading and reconciliation.
Result (what ends it, for whom)	Fuel is delivered to domestic transport, agriculture and industrial customers.
Why this process matters for Company B (one fact + source no.)	Reliable domestic fuel distribution supports transport, agriculture and industry. Source: KMG Annual Report 2025; verify the exact section/page.

Step 2: current state (how it works TODAY)

5–8 steps. Actor = swimlane (customer / staff role / system). Mark the bottleneck row with ● and say how long work waits there.

#	Actor (swimlane)	Step (verb + object)	Time / waiting time	Bottleneck ●	Problem observed
1	B2B customer	Submits order and requested volume	Variable; same day to several days	—	Incomplete order details
2	Regional sales	Checks contract and payment/prepayment status	Hours	—	Manual checks and missing documents

#	Actor (swimlane)	Step (verb + object)	Time / waiting time	Bottleneck ●	Problem observed
3	Planning team	Checks stock and allocation	Several hours	●	Stock visibility may be delayed
4	Refinery operations	Confirms grade and batch availability	Hours	—	Production/quality constraints
5	Quality laboratory	Verifies product specification and certificate	Before dispatch	—	Waiting for test result
6	Rail terminal	Schedules loading slot and prepares railcars	Variable; queue possible	—	Loading-slot congestion
7	Terminal operator	Loads fuel and records meter/quantity data	Hours	—	Meter or data-entry discrepancy
8	Customer / finance	Receives documents and reconciles invoice	After loading	—	Mismatch between meter data and invoice

Step 3: the KPI

KPI name	Definition (how it is calculated)	Baseline today	Source (or 'labelled estimate')	Future-state target
Order-to-loading lead time	Elapsed hours from confirmed order to completed loading; report median and 90th percentile.	48 hours (labelled estimate)	Illustrative only; replace with KMG/terminal records or group data.	Median ≤24 hours; 90% of standard orders within 36 hours.

Step 4: future state (how it SHOULD work — use your Week 4 matrix)

Tag every step with its allocation category: Human Only / Human + AI / AI Executes + Human Review / Automated.

#	Actor (swimlane)	Step (verb + object)	Allocation category	What changed vs current state	Effect on the KPI
1	B2B customer	Submit complete digital order	Automated	Form validates required fields and customer details.	Fewer incomplete orders; shorter lead time.
2	B2B portal	Check contract and payment rules	Automated	Rules-based validation flags exceptions.	Reduces manual review time.
3	Planning system	Check stock and suggest allocation	AI Executes + Human Review	Live stock visibility and demand forecast; planner approves exceptions.	Reduces stock-check wait.
4	Refinery planner	Confirm production/grade availability	Human + AI	AI provides schedule options; planner resolves constraints.	Improves schedule reliability.
5	Quality laboratory	Test sample and approve certificate	Human Only	Human lab sign-off remains mandatory.	Prevents incorrect product release.

#	Actor (swimlane)	Step (verb + object)	Allocation category	What changed vs current state	Effect on the KPI
6	Rail terminal system	Book loading slot and notify customer	Automated	Shared slot calendar reduces duplicate booking.	Reduces queue and waiting.
7	Terminal operator	Load and verify meter reading	Human + AI	Digital meter capture flags abnormal readings; operator confirms.	Reduces data-entry errors.
8	Finance / customer	Reconcile invoice and close order	AI Executes + Human Review	Automated matching flags discrepancies; finance resolves exceptions.	Faster reconciliation; fewer disputes.

Step 5: the controls (required for A2 and rubric row 7)

Control	WHO (named role)	WHAT / WHEN (trigger)	HOW FAST (time limit)
Human review point	**Human review point**	Terminal supervisor / quality manager	Check grade certificate and meter reading before dispatch is finalised.
Escalation path	**Escalation path**	Regional sales manager → terminal manager → operations director	Escalate delay beyond target, stock shortage or repeated loading discrepancy.
Manual fallback (system down / AI wrong)	**Manual fallback (system down / AI wrong)**	Shift supervisor and sales team	If portal/SCADA is unavailable, use approved paper order, manual stock confirmation, phone slot booking and signed meter log; reconcile after restoration.

Five-element check

☒ Bottleneck named and measured in the current state

☒ Every future-state step tagged with a Week 4 allocation category

☒ KPI with baseline and target

☒ Human review point: who checks what, before it counts

☒ Escalation path and manual fallback written out

Worked example (illustrative — a Kazakhstan online seller, not your Company B)

	Current state	Future state
Steps	Order → manager checks stock in Excel twice a day → packs → phones courier → customer calls ‘where is my order?’	Order → stock check (Automated) → pack → courier booked via API (Automated) → SMS tracking (Automated); questions: chatbot (AI Executes + Human Review)
Bottleneck	Manual stock check: orders wait up to 12 hours	Removed; only exceptions go to a human

	Current state	Future state
KPI	Order-to-dispatch time ≈ 26 hours	Under 6 hours; 90% dispatched same day
Human review point	None	Substitutes and refunds over 20,000 KZT approved by the sales manager (Human + AI)
Escalation path	Customer calls whoever answers	Chatbot → human agent after 2 failed replies or the word 'complaint'; agent → manager within 2 hours
Manual fallback	—	System down: printed pick list + courier booked by phone; reconcile within 24 hours

Pull the plug (5 min)

Scenario we received (copy the orange sticky)	Our 60-second answer: who notices, who is called, how work continues	What we fixed in the map
Scenario: B2B portal or terminal data feed is unavailable during peak agricultural demand.	Shift supervisor confirms the outage, informs terminal manager and regional sales, switches to paper orders and phone scheduling, and records loading manually. Reconcile transactions after restoration.	Added a 15-minute fallback trigger, named escalation owners and a 24-hour reconciliation deadline.

A2 defence — practise your answers

The defence is individual, about 5 minutes, without notes or AI. Write short answers here to practise, then say them aloud in under 45 seconds each. Good answer = claim + reason (test or theory) + Company B fact.

Question	My answer (key words only)
My position in one sentence	KMG's fuel order-to-loading process should be digitised to improve visibility and reduce waiting, while quality release and operational exceptions remain human-controlled.
Why is my Human Only task not Human + AI?	Human Only is appropriate for laboratory quality release because an incorrect decision can affect product compliance and safety; AI cannot replace accountable sign-off.
Who checks the AI's work, what, and when?	The quality manager checks the product certificate before release; the terminal operator verifies the meter reading at loading; finance reviews flagged invoice mismatches.
The system fails Monday 09:00 — escalation and fallback?	The shift supervisor identifies the outage, notifies the terminal manager and regional sales manager, and switches to paper orders and manual logs within 15 minutes. Reconcile within 24 hours.
My KPI: number today, source, target	KPI: order-to-loading lead time. Illustrative baseline 48 hours (labelled estimate); target median ≤24 hours and 90% within 36 hours. Replace the estimate with actual terminal data.
Why adapt / pilot / delay / reject, not adopt, Company A's practice?	Adapt Shell's digital customer and logistics practices, but pilot integrations with KMG terminals first because infrastructure, regulation and customer readiness differ.

Weekly Record 5 planner — REDUCED, fields 1–2 only

Field	What goes here	My notes / draft
1 Concept	Key terms in your own words: production planning, process types, JIT, critical path, TQM, lean, bottleneck, KPI. Concept check: "Why does automating a non-bottleneck not increase output?"	P
2 Application	Your current- and future-state maps with bottleneck, KPI, review point, escalation path and manual fallback + 3–4 sentences on what the Paper Airplane Factory taught you.	M
3–5	Not filled separately — absorbed into your A2 Video Essay (script, Evidence Folder, AI-use declaration).	—
Capstone link	Record 5 develops into Capstone section 5a: Operations and supply chain.	—